Percolation in a One–Dimensional Lattice

1 Site Percolation in 1D

Consider a one–dimensional lattice containing N sites.
Each site can be

- occupied with probability p
- empty with probability $1 - p$

The occupation of different sites is independent.
A **cluster** is defined as a group of consecutive occupied sites.
For example, if three neighbouring sites are occupied, they form a cluster of size $s = 3$.

2 Example Configuration

○ ● ● ○ ○ ● ● ● ○ ○ ● ○ ○

$s = 2$ $s = 3$ $s = 1$

Filled circles represent occupied sites. Consecutive filled sites form clusters.

3 Cluster Size Distribution

Let

$$N_s$$

denote the total number of clusters of size s in the lattice.
Since the lattice contains N sites, it is convenient to define the **cluster density**

$$n_s = \frac{N_s}{N}$$

which represents the **number of clusters of size s per lattice site**.

3.1 Why divide by N?

If we only used N_s, the number of clusters would depend on the system size.
By dividing by N, we obtain a quantity that remains finite in the **thermodynamic limit**

$$N \to \infty$$

Thus

$$n_s = \frac{N_s}{N}$$

is an intensive quantity that characterizes the statistical properties of the lattice. This definition is analogous to defining densities such as

$$\rho = \frac{N}{V}$$

in statistical mechanics.

4 Probability for a Cluster of Size s

To form a cluster of size s:

- s consecutive sites must be occupied
- the site before the cluster must be empty
- the site after the cluster must be empty

Therefore the probability of observing such a configuration is

$$n_s = p^s(1-p)^2$$

Here

- p^s ensures s occupied sites
- $(1-p)^2$ ensures the cluster is bounded by empty sites

5 Fraction of Occupied Sites

The number of sites belonging to clusters of size s is

$$sN_s$$

Dividing by N gives the fraction of sites belonging to clusters of size s

$$\frac{sN_s}{N} = sn_s$$

Summing over all cluster sizes gives the total occupied fraction

$$f = \sum_{s=1}^{\infty} sn_s$$

Substituting n_s

$$f = \sum_{s=1}^{\infty} sp^s(1-p)^2$$

We factor out $(1-p)^2$

$$f = (1-p)^2 \sum_{s=1}^{\infty} s p^s$$

Now we use the geometric series

$$\sum_{s=1}^{\infty} p^s = \frac{p}{1-p}$$

Differentiating with respect to p

$$\frac{d}{dp}\left(\sum p^s\right) = \sum s p^{s-1}$$

Multiplying by p

$$\sum s p^s = p \frac{d}{dp}\left(\frac{p}{1-p}\right)$$

Now

$$\frac{d}{dp}\left(\frac{p}{1-p}\right) = \frac{1}{(1-p)^2}$$

Thus

$$\sum s p^s = \frac{p}{(1-p)^2}$$

Substituting back

$$f = (1-p)^2 \frac{p}{(1-p)^2}$$

Hence

$$f = p$$

which is consistent with the occupation probability.

6 Mean Cluster Size

Define the probability that a randomly chosen occupied site belongs to a cluster of size s

$$W_s = \frac{s n_s}{\sum s n_s}.$$

The mean cluster size is

$$\langle s \rangle = \sum s W_s.$$

Note:

If we choose an occupied site at random, the probability that it belongs to a cluster of size s must be proportional to the number of occupied sites contained in clusters of that size. Normalizing by the total fraction of occupied sites $\sum_s s n_s$, we obtain

$$W_s = \frac{sn_s}{\sum_s sn_s}.$$

Thus W_s represents the probability that a randomly chosen occupied site belongs to a cluster of size s. The mean cluster size is therefore

$$\langle s \rangle = \sum_s sW_s.$$

To compute such quantities systematically, it is convenient to introduce the *cluster generating function*, which encodes the entire cluster size distribution and allows the moments of the distribution to be obtained by differentiation.

Substituting W_s

$$\langle s \rangle = \sum s \frac{sn_s}{\sum sn_s}$$

$$\langle s \rangle = \frac{\sum s^2 n_s}{\sum sn_s}.$$

Since

$$\sum sn_s = p$$

we obtain

$$\langle s \rangle = \frac{1}{p} \sum s^2 n_s$$

Now

$$\sum s^2 n_s = \sum s^2 p^s (1-p)^2$$

Factor $(1-p)^2$

$$= (1-p)^2 \sum s^2 p^s$$

Using

$$\sum s^2 p^s = \frac{p(1+p)}{(1-p)^3}$$

we obtain

$$\sum s^2 n_s = (1-p)^2 \frac{p(1+p)}{(1-p)^3}$$

$$= \frac{p(1+p)}{1-p}$$

Thus

$$\langle s \rangle = \frac{1}{p} \frac{p(1+p)}{1-p}$$

Hence

$$\boxed{\langle s\rangle = \frac{1+p}{1-p}}$$

7 Characteristic Cluster Size

The cluster distribution is

$$n_s = p^s(1-p)^2$$

Rewrite

$$p^s = e^{s\ln p}$$

Thus

$$n_s = (1-p)^2 e^{s\ln p}$$

Define a characteristic cluster size s_ξ such that

$$n_s \sim e^{-s/s_\xi}$$

Comparing exponents gives

$$-\frac{1}{s_\xi} = \ln p$$

Therefore

$$\boxed{s_\xi = -\frac{1}{\ln p}}$$

8 Behaviour Near the Critical Point

For one–dimensional percolation

$$p_c = 1$$

Let

$$x = p_c - p$$

Since $p_c = 1$

$$p = 1 - x$$

For small x

$$\ln p = \ln(1-x)$$

Using the expansion

$$\ln(1-x) \approx -x$$

Thus

$$s_\xi = -\frac{1}{\ln p} \approx \frac{1}{x}$$

Hence

$$\boxed{s_\xi \sim (p_c - p)^{-1}}$$

9 Critical Behaviour of Mean Cluster Size

Using

$$\langle s \rangle = \frac{1+p}{1-p}$$

near $p_c = 1$

$$\langle s \rangle \approx \frac{2}{1-p}$$

Thus

$$\boxed{\langle s \rangle \sim (p_c - p)^{-1}}$$

Therefore the mean cluster size diverges as the critical point is approached.

10 Evaluating the Sums Using the Operator Method

In the derivation of the mean cluster size we encounter sums of the form

$$\sum_{s=1}^{\infty} s p^s \qquad \text{and} \qquad \sum_{s=1}^{\infty} s^2 p^s$$

A convenient method to evaluate such sums is to use the operator

$$p\frac{d}{dp}$$

which acts as a generator of powers of s.

10.1 Basic Geometric Series

We begin with the geometric series

$$\sum_{s=1}^{\infty} p^s = \frac{p}{1-p}$$

for $|p| < 1$.

10.2 Generating the Factor s

Consider the operator

$$p\frac{d}{dp}$$

Acting on p^s gives

$$p\frac{d}{dp}(p^s) = p(sp^{s-1}) = sp^s$$

Thus

$$p\frac{d}{dp}\left(\sum p^s\right) = \sum sp^s$$

Now apply the operator to the geometric series result

$$\sum p^s = \frac{p}{1-p}$$

Therefore

$$\sum sp^s = p\frac{d}{dp}\left(\frac{p}{1-p}\right)$$

Now compute the derivative

$$\frac{d}{dp}\left(\frac{p}{1-p}\right) = \frac{1}{(1-p)^2}$$

Hence

$$\boxed{\sum sp^s = \frac{p}{(1-p)^2}}$$

10.3 Generating the Factor s^2

Applying the operator twice produces s^2

$$\left(p\frac{d}{dp}\right)^2 p^s = s^2 p^s$$

Thus

$$\sum s^2 p^s = \left(p\frac{d}{dp}\right)^2 \sum p^s$$

Using the geometric series again

$$\sum p^s = \frac{p}{1-p}$$

First application:

$$p\frac{d}{dp}\left(\frac{p}{1-p}\right) = \frac{p}{(1-p)^2}$$

Second application:

$$p\frac{d}{dp}\left(\frac{p}{(1-p)^2}\right)$$

Compute the derivative

$$\frac{d}{dp}\left(\frac{p}{(1-p)^2}\right) = \frac{1+p}{(1-p)^3}$$

Thus

$$\boxed{\sum_{s=1}^{\infty} s^2 p^s = \frac{p(1+p)}{(1-p)^3}}$$

10.4 Application to Percolation

Recall that

$$n_s = p^s(1-p)^2$$

Therefore

$$\sum s^2 n_s = (1-p)^2 \sum s^2 p^s$$

Substituting the result

$$\sum s^2 n_s = (1-p)^2 \frac{p(1+p)}{(1-p)^3}$$

which simplifies to

$$\sum s^2 n_s = \frac{p(1+p)}{1-p}$$

Substituting into the mean cluster size

$$\langle s \rangle = \frac{1}{p} \sum s^2 n_s$$

we obtain

$$\boxed{\langle s \rangle = \frac{1+p}{1-p}}$$

This operator method provides a systematic way to evaluate cluster moment sums in percolation theory.

11 Additional remarks- out of syllabus

11.1 Generating Function for One–Dimensional Percolation

For site percolation in one dimension the cluster distribution is

$$n_s = p^s(1-p)^2.$$

Substituting this expression into the definition of the generating function,

$$G(z) = \sum_{s=1}^{\infty} n_s z^s,$$

we obtain

$$G(z) = \sum_{s=1}^{\infty} p^s(1-p)^2 z^s.$$

Factor out $(1-p)^2$,

$$G(z) = (1-p)^2 \sum_{s=1}^{\infty} (pz)^s.$$

The remaining sum is a geometric series,

$$\sum_{s=1}^{\infty} (pz)^s = \frac{pz}{1-pz},$$

valid for $|pz| < 1$. Therefore the generating function becomes

$$\boxed{G(z) = (1-p)^2 \frac{pz}{1-pz}}$$

which completely characterizes the cluster size distribution in one dimension.

11.2 Recovering Physical Quantities from $G(z)$

The generating function allows us to compute moments of the cluster distribution.

The first derivative is

$$G'(z) = \sum_{s=1}^{\infty} s n_s z^{s-1}.$$

Evaluating at $z = 1$ gives

$$G'(1) = \sum s n_s.$$

Using the explicit form of $G(z)$,

$$G(z) = (1-p)^2 \frac{pz}{1-pz},$$

we differentiate,

$$G'(z) = (1-p)^2 \frac{p(1-pz) + p^2 z}{(1-pz)^2}.$$

Setting $z = 1$,

$$G'(1) = (1-p)^2 \frac{p}{(1-p)^2}.$$

Thus

$$\sum s n_s = p.$$

This reproduces the expected result that the fraction of occupied sites is p.

11.3 Mean Cluster Size from the Generating Function

Recall that

$$\langle s \rangle = \frac{\sum s^2 n_s}{\sum s n_s}.$$

Using the generating function,

$$\sum s^2 n_s = G''(1) + G'(1).$$

Carrying out the differentiation and simplifying gives

$$\sum s^2 n_s = \frac{p(1+p)}{1-p}.$$

Substituting into the expression for the mean cluster size,

$$\langle s \rangle = \frac{1}{p} \sum s^2 n_s,$$

we obtain

$$\boxed{\langle s \rangle = \frac{1+p}{1-p}.}$$

Thus the generating function provides a compact and systematic way to compute the moments of the cluster size distribution.

11.4 Generating Function and Critical Behaviour

The generating function also provides insight into the critical behaviour of the system. Recall that

$$G(z) = (1-p)^2 \frac{pz}{1-pz}.$$

This function has a pole at

$$z = \frac{1}{p}.$$

The position of this singularity determines the large–s behaviour of the cluster distribution. Expanding $G(z)$ as a power series in z yields

$$n_s \sim p^s.$$

Thus the cluster distribution decays exponentially with cluster size.

This exponential form can be written as

$$n_s \sim e^{-s/s_\xi},$$

where the characteristic cluster size is

$$s_\xi = -\frac{1}{\ln p}.$$

As the occupation probability approaches the critical point

$$p_c = 1,$$

we have

$$\ln p \approx -(1-p),$$

and therefore

$$s_\xi \sim (p_c - p)^{-1}.$$

Thus the characteristic cluster size diverges as the critical point is approached. This divergence signals the onset of long-range connectivity in the system and is the hallmark of critical behaviour in percolation.